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Vibrissae-inspired wearable neuromorphic artificial fingernail for vibrotactile perception and interfacing

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Abstract

The rodent vibrissal system exhibits remarkable vibrotactile perception, enabling the discrimination of fine surface features through dynamic spike encoding and efficient synaptic integration. Here, we develop a vibrissae-inspired wearable neuromorphic artificial fingernail that combines piezoelectric sensing, spike encoding, and synaptic processing within a near-sensor computing architecture. A dual-gate synaptic transistor gated by proton-conducting ion-gel exhibits excellent air stability, sensory memory, sensory integration, and perceptual weighting capabilities, allowing spatiotemporal recognition of both fast- and slow-adapting tactile spikes. Notably, this synaptic device features multiple preprocessing functions, enabling direct extraction of task-relevant features at the hardware level and thereby minimizing computational requirements. Using this system, we achieve over 90% classification accuracy across multiple tactile tasks, including object contour recognition, surface texture discrimination, and finger-writing trajectory identification. These results demonstrate the potential of our system as a smart wearable interface for low-power, real-time tactile recognition, thereby expanding the applications of neuromorphic electronics.

Keywords

neuromorphic tactile system, vibrotactile perception, near-sensor computing, wearable electronics, human–machine interaction

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1. Introduction

Tactile perception is a critical function of biological sensory systems, playing a vital role in object recognition, environmental exploration, and physical interaction with the surrounding world[1-3]. Biomimetic approaches to tactile sensing [4-5] have demonstrated significant advancements in replicating complex sensory functions of biological skin, resulting in artificial tactile systems such as electronic skin that show promise in applications including robotics, prosthetics, and human–machine interfaces. Compared with traditional skin-based tactile perception, the whisker-based sensory system in rodents exhibits superior sensitivity, spatial resolution, and dynamic responsiveness[6-7]. This capability is believed to stem from the spike encoding scheme and early-stage synaptic integration[8-9]. The former, comprising fast- (FA) and slow-adapting (SA) mechanoreceptors, enables event-driven sensing by selectively converting dynamic and static tactile inputs into sparse spike trains, consuming energy only when activated by relevant mechanical stimuli[10-12]. The latter further integrates these spikes at the neuronal level, allowing essential spatiotemporal features to be extracted before reaching higher-order processing centers[13-14]. Despite the significant attention given to the remarkable capabilities of rodent whisker sensing, several challenges remain in replicating this biological mechanism in artificial tactile systems[15-16]. Most current approaches focus primarily on sensor structures and material properties, while still relying on complex algorithms or external processing units for feature extraction and classification[17-19]. These strategies introduce significant hardware complexity and computational load, which hinder their use in wearable applications. Therefore, there remains a critical need for a fully integrated, energy-efficient, real-time tactile system inspired by rodent whiskers.

In this work, we report a neuromorphic fingernail sensory system designed for vibrotactile perception and tactile interaction. The fingernail-mounted flexible sensor detects pressure and vibration, while also providing a direct tactile interface for human–machine interactions. The system adopts the connection architecture and encoding strategy of biological mechanoreceptors, where FA and SA pathways jointly encode dynamic and static tactile information. The dual-gate indium-gallium-zinc-oxide (IGZO) synaptic transistor, gated by a proton-conducting ion-gel, serves as the synaptic processing unit. It exhibits synaptic plasticity with asymmetric weight, enabling spatiotemporal integration and sensory weighting between FA and SA channels. Various tactile perception tasks, including vibration pattern recognition, contour and texture discrimination, and finger-writing trajectory recognition, are performed to validate the system's capability for tactile cognition and human–machine interaction in real-world settings. Unlike electronic-skin systems that depend on dense sensor arrays and external computing resources, our neuromorphic fingernail performs hardware-level spike encoding and sensory integration within a compact, wearable platform. This architecture enables real-time, low-power tactile recognition, and it also endows wearable interactive devices with bioinspired perceptual capability and neuromorphic intelligence.

2. Experimental section

2.1 Fabrication of synaptic devices

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Dual-gate IGZO synaptic transistors were fabricated on 1.5 cm ×1.5 cm Si substrates (Figure S1). Substrates were ultrasonically cleaned in acetone, isopropanol, and deionized water, followed by ultraviolet-ozone (UV-ozone) treatment to improve surface cleanliness and wettability. A 30 nm-thick IGZO (In:Ga:Zn = 2:2:1) semiconductor layer was deposited by radio frequency (RF) magnetron sputtering (100 W, 5 mTorr Ar atmosphere, room temperature). The source, drain, and two lateral gate electrodes were defined through a Ni shadow mask, followed by the sputtering of a 100 nm indium tin oxide (ITO, mass ratio of In_2O_3 and SnO_2 was 9:1) film under identical Ar pressure. The chitosan-based ion-gel electrolyte was prepared by dissolving chitosan (2 wt%) in acetic acid (2 wt%) at 80 °C with continuous stirring, followed by the addition of 1 wt% glycerol. The resulting solution was cast and thermally cured at 80 °C for 1 h to form a flexible, transparent gel film. The ion-gel film was then diced and laminated onto the surface of the oxide channel and planar gates, establishing an electric double layer at the gel/semiconductor interface. The obtained devices were wire-bonded onto printed circuit boards for testing. The structural morphology and layer configuration of the as-fabricated device were characterized by optical microscopy, cross-sectional scanning electron microscopy (SEM), and atomic force microscopy (AFM), as provided in Figure S2. The device architecture supports single-gate or dual-gate operation, with synaptic plasticity modulated by electrical spikes.

2.2 Characterization and measurements

The structural morphology and layer configuration of the artificial synaptic device were examined by optical microscopy (DM500, Leica), cross-sectional field-emission scanning electron microscopy (SEM, ULTRA 55, Carl Zeiss), and atomic force microscopy (AFM, MultiMode 8, Bruker Scientific Instruments). The ionic capacitance of the ion-gel was evaluated using an LCR meter (Agilent E4980A). The synaptic transistor was electrically characterized using a semiconductor parameter analyzer (Keysight B1500A). To characterize the piezoelectric sensor response, mechanical stimulation was applied using a motorized force test stand (ESM301, Mark-10) equipped with a force gauge, and the resulting voltage signals were recorded with a Tektronix 4 Series MSO oscilloscope (MSO44).

2.3 Hardware of the neuromorphic system

A flexible piezoelectric vibrotactile sensor (LDT0-028K, TE Connectivity) was employed to transduce mechanical deflection into analog voltage signals. The sensor, composed of a 28 μm-thick piezoelectric film bonded to a polyester substrate with screen-printed electrodes, was mounted onto a custom-designed printed circuit board (PCB). The PCB provided a direct connection to the dual-gate synaptic transistor via a gold-plated interface for real-time signal transmission. The spike-encoded FA/SA signals were routed to the two independent gates of the transistor, which was powered by a 0.1 V supply module controlled by the microcontroller (ESP32). ESP32 also read the drain-source current through an integrated analog front-end module equipped with a low-noise transimpedance amplifier and analog-to-digital converter (ADC) circuitry, enabling precise capture of the postsynaptic current (PSC) (Figure S3). The difference between the maximum and minimum postsynaptic current, denoted as ΔPSC, was extracted to quantify the stimulus intensity. Hardware-derived features, including f_{FA} and f_{SA}, and ΔPSC (Figure S4), were processed by the microcontroller for real-time tactile classification. All feature data and recognition results were wirelessly

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transmitted via Bluetooth to a host computer (Figure S5). This wireless, distributed sensing-processing architecture supports compact, low-latency, and wearable closed-loop tactile cognition without reliance on external computation units.

2.4 Demonstration of tactile perception and finger-writing interaction

All the experiments were performed using a biomimetic tactile perception system consisting of a piezoelectric sensor, a spike-encoding module, and a dual-gate IGZO synaptic transistor. To minimize power-line interference, all sensor signal acquisitions were conducted inside an electromagnetic shielding cage. For tactile scanning tasks, the piezoelectric sensor was mounted on a linear translation stage (Zolix TSM25-2A) and inclined by ~10° relative to the scanning plane to ensure reliable contact. A pre-deflection of ~1 mm was applied to the sensor tip to ensure stable contact with the sample surface. The sample was translated at a constant speed of 5.25 $mm \cdot s^{-1}$. Sensor output signals were acquired using a Tektronix 4-series oscilloscope and processed in real time via a spike encoding algorithm. Encoded spikes were sent to the dual-gate terminals of the IGZO-based synaptic transistor, and the corresponding postsynaptic current signals from the device were recorded. Encoded spikes, postsynaptic current, and classification outcomes were wirelessly transmitted via Bluetooth and visualized on a host computer for analysis.

3. Results and discussion

3.1 Bioinspired artificial fingernail with neuromorphic architecture

In the rodent vibrissal system, when a vibrissa contacts external objects, mechanical stimuli such as vibration and pressure are captured by the follicle-sinus complex located at the vibrissae base (Figure 1(a))[10, 20]. This specialized tactile organ contains diverse mechanoreceptors with different sensitivities to stimulus dynamics[10]. Some receptors respond preferentially to rapid, transient deflections with phasic firing, whereas others exhibit more sustained responses with tonic firing[21]. These two distinct spike patterns are conveyed by different classes of afferent fibers to second-order neurons in the trigeminal nuclei, where synaptic integration occurs[22]. This early-stage processing emphasizes stimulus-dependent temporal dynamics and reduces redundancy in the tactile inputs before transmission to the somatosensory cortex for further perception and recognition[23]. The entire process can be conceptually understood through a bioinspired labeled-line framework (Figure S6), in which dynamic and static tactile cues are independently encoded through two parallel channels (FA and SA) and subsequently integrated at a later stage.

Figure 1(b) shows the schematic of our wearable neuromorphic artificial fingernail, which reproduces key structural and functional elements of the rodent vibrissae pathway for sensing, encoding, and preprocessing. Our system employs a flexible piezoelectric sensor that transduces mechanical stimuli such as vibration and pressure into continuous analog voltage signals resembling the receptor potential[24]. We designed a dual-channel spike encoder that converts analog tactile signals into two spike trains corresponding to dynamic (FA) and static (SA) tactile information. These two spike trains are delivered to the two gates of a dual-gate synaptic transistor with asymmetric synaptic weights, reflecting the different perceptual weights among multiple tactile modalities in biological systems. The temporal pattern and correlation of the two spike

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trains modulate the synaptic behavior of the transistor and consequently affect its postsynaptic current output. In this way, the device realizes multisensory integration and synaptic processing of tactile stimuli, analogous to mechanoreceptor pathways in biological tactile perception[25-26].

Standard machine learning (ML) pipelines for tactile recognition typically require the preprocessing of temporal signals, which are then unfolded into high-dimensional feature vectors and processed by computationally intensive models such as deep neural networks, spiking neural networks, or other high-dimensional classifiers (left panel of Figure 1(c)) [27-29]. Instead of expanding temporal signals into high-dimensional representations, our strategy (right panel of Figure 1(c)) encodes tactile information at the sensing stage by converting continuous tactile signals into sparse, event-driven spike trains. These spikes are then processed by a dual-gate synaptic transistor to perform weighted integration of the FA and SA pathways, thereby producing synaptic responses that reflect their spike-timing dynamics. The synaptic responses of the device together with the mean firing rate of the FA and SA spikes form a low-dimensional feature set, which can be directly analyzed by lightweight models for energy-efficient tactile recognition.

3.2 Neuromorphic processing workflow of the bioinspired tactile system

The overall workflow of our bioinspired tactile system is illustrated in Figure 2(a). Tactile inputs are encoded into spike trains, processed by a synaptic transistor through integration and nonlinear modulation, and finally reduced to compact features for recognition. This architecture clearly separates sensing, encoding, preprocessing, and recognition, emulating the organization of biological vibrissal pathways. At the tactile sensing stage (Figure 2(b)), a piezoelectric fingernail serving as the primary transducer generates signals that preserve the fine temporal features of the tactile stimuli. Three types of tactile stimuli are explored, including vibration, pressure, and texture, corresponding to FA, SA, and combined FA/SA responses. To actively explore objects, the artificial fingernail is mounted on a human finger. The piezoelectric sensor on the artificial fingernail can simultaneously capture both dynamic vibration and static deformations with high sensitivity. The soft piezoelectric sensor conforms to curved surfaces, maintains intimate contact during scanning, and is highly sensitive to subtle vibrations. In particular, it can directly capture vibrotactile signals from a vibrating motor with high temporal fidelity (Figure S7). The continuous analog signals are then transformed into sparse, event-driven spike trains at the encoding stage (Figure 2(c)). FA spikes are generated when the change rate of the signal exceeds a predefined threshold, thereby capturing transient, dynamic events. The SA spikes are elicited when the absolute amplitude of the signal exceeds a distinct threshold, effectively encoding sustained pressure stimuli (Note S1). The main encoding parameters and their values are summarized in Table S1, and the influence of the encoding thresholds on recognition performance is further analyzed in Figure S8. The underlying principle of this approach is the representation of vibrotactile stimuli using spatiotemporal patterns of spike trains, where the spiking dynamics encode both the temporal and spatial features of the stimulus. This spike encoding allows efficient capture of dynamic tactile information, reflecting the key characteristics of vibration and pressure through the precise timing and frequency of the spikes.

Synaptic preprocessing (Figure 2(d)) is implemented by a dual-gate synaptic transistor with an asymmetric synaptic weight, which achieves multiple preprocessing functions. Standard preprocessing

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requires multiple discrete components, such as amplifiers, filters, and nonlinear circuits (Note S2). In contrast, our dual-gate synaptic transistor consolidates these functions within a single device. This significantly reduces both the hardware complexity and associated power consumption. A detailed quantitative analysis of energy consumption and system-level power breakdown is provided in Figures S9 and S10, Tables S2 and S3. The results indicate that the event-level energy consumption remains on the order of several hundred nanojoules, while the total system power is maintained within the microwatt range. Such a low power level is compatible with operation using a small lithium battery, supporting the practical requirements of wearable electronics. We fabricated a dual-gate synaptic transistor based on chitosan electrolyte. The fabrication process is compatible with flexible substrates, and the device maintains stable performance on polyimide films (Figure S11). Mechanical reliability was evaluated under repeated deformation. The bending radius and retention of synaptic characteristics after repeated bending cycles are shown in Figure S12, while the AFM surface morphology before and after bending is presented in Figure S13, confirming mechanical robustness of the device. Scalable integration of ion-gel arrays can be achieved via inkjet printing[30], enabling large-area flexible electronics. To evaluate the device-to-device variation, we statistically analyzed 50 devices from the same batch (Figure S14). Both the dual-gate synaptic responses and the asymmetric weighting ratio remain within a narrow range, indicating good uniformity.

Synaptic functionality arises from proton-mediated lateral gating. Presynaptic positive pulses drive protons in the chitosan electrolyte to accumulate at the chitosan/IGZO interface, forming an electric double layer (EDL)[31]. The formation and relaxation of this EDL govern short-term plasticity, enabling transient memory effects analogous to biological synapses[32]. A detailed analysis of the proton-mediated mechanism, including relaxation kinetics and frequency-dependent capacitance, is provided in Figures S15 and S16. The device exhibits a pronounced hysteresis window (~4 V) in its transfer curve, corresponding to its memory behavior (Figure S17). The postsynaptic current (PSC) was read from the drain current and regulated by voltage spikes applied to Gate1. Under positive presynaptic stimulation of 6 V with a pulse width of 50 ms, the device exhibited an excitatory response, which was further verified by the paired-pulse facilitation (PPF) behavior shown in Figure S18. As either the spike number or spike frequency increased, the PSC progressively increased, demonstrating spike-number-dependent plasticity (SNDP; Figure S15(a)) and spike-rate-dependent plasticity (SRDP; Figure S15(b)). By increasing the pulse width, the peak amplitude of the PSC also increased nearly linearly (SDDP; Figure S15(c)), indicating that the synaptic strength can be tuned by the duty cycle of presynaptic spikes[33]. These stimulus-dependent synaptic behaviors are consistent with proton accumulation and relaxation at the ion-gel/IGZO interface under EDL gating.

The memory retention behavior of the device can be further modulated by different sustained spikes, with the device transitioning from short-term to long-term memory under fixed biasing (Figure S19). Cycling stability was evaluated by monitoring the synaptic response under repeated pulse operation (Figure S20(a)). The device retained stable synaptic behavior over 1 000 operation cycles, with an overall PSC degradation of less than 5%. Environmental stability was tested under high-temperature and high-humidity condition (Figure S21). After storage at 40 °C and 80% RH for 3 days, the device retained a clear hysteresis window and stable

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synaptic response, with variations in the PSC peak and a relaxation time below 5%. Such ambient stability suggests the potential for wearable electronics applications.

Due to the asymmetric configuration, the two gates have different effective coupling distances to the channel. The gate near the drain has a shorter coupling distance, accumulates more protons, and thus produces a stronger PSC. This asymmetry assigns greater weight to the FA input and lower weight to the SA input, mimicking the different contributions of FA and SA receptors to tactile perception in biological systems. When identical spikes are applied, the response from the drain-side gate is nearly twice as large as that from the source-side gate. A quantitative comparison of the dual-gate synaptic weights under different stimulation conditions is provided in Figure S22. The response of a control device with approximately equivalent gate-coupling geometry further supports that the asymmetric weighting is governed by the difference in the gate-coupling condition (Figure S23). This asymmetric dual-gate design enables FA and SA spikes to be processed with different synaptic weights. The effect of the FA/SA weighting ratio on recognition performance is shown in Figure S24. Notably, the corresponding FA/SA weighting ratio remained nearly unchanged during repeated operations (Figure S20(b)), confirming the stability of the asymmetric weighting function that underlies dual-channel tactile processing.

The asymmetric weighting further enables spatiotemporal processing of paired-spike trains. When two spike trains are applied sequentially to different gates, the resulting PSC depends on both the inter-spike interval and the order of arrival. Exchanging the sequence of inputs produces clearly distinguishable responses. The second pulse evokes a larger PSC than the first pulse because of residual proton accumulation, and the relative amplitude strongly depends on which gate is stimulated first. Measurements under different inter-spike intervals show variations in PSC, confirming the spatiotemporal recognition capability. This spatiotemporal selectivity enables the device to perform real-time sensory integration and processing at the hardware level.

The processed signals are ultimately translated into classification outcomes at the recognition stage (Figure 2(e)). In biological somatosensory systems, tactile perception depends not only on instantaneous firing but also on time-averaged neuronal responses[34]. Distinct sensory neurons, such as FA and SA mechanoreceptors, are characterized by their mean firing rates and discharge patterns, which together define the temporal coding of tactile stimuli[25]. Inspired by this neural principle, we adopted a time-averaging strategy to quantify spike-encoding behavior. For each tactile event, the stimulus duration defines the averaging window. The mean firing rates (f_{FA}, f_{SA}) of the FA and SA channels are calculated by dividing the total spike count by the event duration. The change in postsynaptic current (ΔPSC) represents the postsynaptic response of the device. The firing rates serve as presynaptic indicators of temporal activity, reflecting dynamic and static tactile information. The firing rate alone cannot be used to distinguish the sequential order or temporal correlation between the two spike trains (Figure S25). In contrast, the postsynaptic response encodes both the relative timing and strength of the paired inputs, enabling the discrimination of different spatiotemporal patterns. These features are obtained directly at the hardware level from the spike encoder and synaptic device. The resulting feature space exhibits clear separability, with vibration-, pressure-, and texture-

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related information occupying distinct regions. We compared our hardware-based approach against several software-based methods, including deep neural networks (DNNs)[35], spiking neural networks (SNNs)[36], and support vector machines (SVMs)[37] (Note S3). Metrics such as input dimensionality and floating-point operations (FLOPs) per inference show that our system reduces computational cost by orders of magnitude while maintaining high recognition accuracy and task adaptability. Compared with other neuromorphic tactile systems, our strategy achieves low computational complexity with competitive performance (Table S4).

3.3 Multi-task tactile perception and classification

Rodents rely on active vibrissae exploration to perceive and discriminate environmental cues.[38-40] Similarly, our system accomplishes various tasks with high accuracy through near-sensor computing (Figure 3(a)). We demonstrate the tactile perception capability of our neuromorphic fingernail in dynamic vibration pattern recognition and static tactile discrimination (Figure 3(b)). For the vibration tasks, the neuromorphic fingernail was gently pressed against a smartphone screen. For the contour and texture tasks, it was positioned at a fixed tilt angle relative to the scanned surface and moved at constant speed, mimicking natural tactile exploration. The device was reset by applying a negative spike when the system detected no tactile input for 3.0 s.

To evaluate dynamic tactile cognition, we conducted a smartphone vibration recognition task, typical vibrotactile scenario in daily interactions. Five vibration modes, namely, alert, heartbeat, rapid, S.O.S., and symphony (denoted as R1–R5), were tested, each of which exhibited distinct temporal profiles and durations (Figure S26). All tactile events were mapped into a unified feature space defined by the mean firing rates of the FA and SA channels (f_{FA}, f_{SA}) and the postsynaptic current change (ΔPSC). For vibratory inputs, rapid oscillations triggered mainly transient FA responses, while the short duration of each cycle was insufficient to surpass the threshold for sustained SA activation. The corresponding spike trains and postsynaptic current responses (Figure 3(c) and (d)) were mapped onto a two-dimensional feature space defined by the firing rate and postsynaptic current change (Figure 3(e)). Using a spatial clustering algorithm to derive decision boundaries (Note S4), the five vibration modes were successfully classified with an overall accuracy of 96% (Figure 3(f)). The experimental setup and measurement protocol for smartphone vibration recognition are provided in Figure S27 and Note S5. These results demonstrate that dynamic tactile events can be recognized through intrinsic temporal encoding and synaptic integration.

We also examined the system response to sustained tactile inputs in a contour recognition experiment. By scanning across the surface of different objects, low-frequency deformation signals evoke persistent SA firing, analogous to the slowly adapting mechanoreceptor responses (Figure S28). The corresponding SA spike trains and PSC curves for five distinct contours are shown in Figure 3(g) and (h). In contrast to the transient dynamics of vibration stimuli, these sustained responses reflect the spatial continuity and curvature of the contacted object. The resulting features, defined by f_{SA} and ΔPSC, form distinct clusters corresponding to different contours (Figure 3(i)), and the confusion matrix (Figure 3(j)) demonstrates reliable recognition performance.

We further validated the system's capability for complex tactile perception by performing texture discrimination tasks. Scanning across a small bump (Figure S29) reveals a characteristic sequence of contact,

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sliding, release, and relaxation. Afterward, we selected five representative plant leaves (cherry leaf, trillium leaf, chestnut leaf, hazelnut leaf, and magnolia leaf; denoted as L1–L5) with naturally occurring textures to evaluate their tactile discrimination ability in real-world settings. The original piezoelectric signals exhibit diverse dynamic and static features (Figure S30), and the resulting spike trains and PSC curves are shown in Figure 3(k) and (l). In this case, the feature distribution spans all three axes (f_{FA}, f_{SA}, and ΔPSC), and the inclusion of ΔPSC as a third feature enhances separability (Figure 3(m)). The confusion matrix (Figure 3(n)) shows robust recognition accuracy across all categories.

In addition, we evaluated the recognition performance using a lightweight decision-tree classifier as an alternative, and the results for the three representative tasks are presented in Figure S31. Considering that the contact angle and scanning speed can affect the temporal characteristics of the encoded spikes, additional experiments were performed under various scanning conditions. The results (Figures S32–S35) show that the system maintains robust recognition performance across different active exploration conditions. We further evaluated the system's response to more delicate surface patterns, such as fabric textures (Figure S36), where millimeter-sized features can be well discerned using a single sensor. Unlike the above tasks, Braille recognition relies on the spatial arrangement of raised dots in a 2×3 matrix, which cannot be fully resolved by a single sensor. Therefore, we employed two sensors in parallel, and consequently Braille letters were successfully recognized (Figure S37). This result demonstrates that our system can be readily extended to multi-sensor configurations for spatially complex tasks. Moreover, Figures S38 and S39 present a preliminary crosstalk evaluation for one sensing channel under isolated operation and under interfered operation during simultaneous parallel operation.

The dimensionality reduction and feature extraction in our system stem directly from biological principles of sensory processing, where high-dimensional mechanoreceptor signals are condensed into compact neural codes. By mapping continuous tactile waveforms into three biologically interpretable quantities (mean firing rates of FA and SA channels and the postsynaptic current change), the system emulates the efficient sensory processing mechanism in somatosensory pathways. Practically, these low-dimensional features can be implemented in a microcontroller with spike counting functions and readout circuit, allowing real-time, energy-efficient recognition in wearable electronics.

3.4 Finger-writing interface for real-time recognition and control

Rodent whiskers play an important role in active tactile exploration. Whisker-to-whisker and whisker-to-body contacts provide mechanosensory cues for spatial awareness and proximity detection (Figure 4(a))[41,42]. Inspired by this, we developed a neuromorphic human–machine interface (HCI) that interprets finger-writing patterns as device-control commands. Literature works on hand-based assistive interfaces often relied on external sensors (e.g., cameras or inertial measurement units) [43] or continuous visual feedback[44]. In contrast, our system uses a finger-mounted piezoelectric sensor to detect tactile stimuli on the palm as the input command (Figure 4(b)). Finger-writing motion can be decomposed into distinct tactile phases, including initial contact, sliding, lift-off, and release (Figure S40). The FA pathway captures rapid fluctuations during contact and lift-off, whereas the SA pathway records sustained pressure during sliding. Five finger-writing patterns,

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including dot, stroke, triangle, circle, and square, were tested. The piezoelectric sensor was fixed on the fingertip of the right hand, and the commands were written on the palm of the left hand (Figure S41). The FA and SA spike trains (Figure 4(c)) encoded from the sensor signal were processed by the dual-gate synaptic transistor to generate distinct PSC responses for each command (Figure 4(d)). The extracted features, including the mean firing rates of the FA and SA channels, exhibited partially overlapping distributions in the two-dimensional feature plane (Figure 4(e)), reflecting the similarity in tactile dynamics. Adding ΔPSC as a third dimension to the feature space significantly enhanced separability (Figure 4(f)). The classification results summarized in the confusion matrix (Figure 4(g)) show an overall accuracy exceeding 90%, confirming that spike-based encoding and hardware-based processing enable reliable recognition of finger-written inputs.

Finally, we implemented haptic control of a smart lamp, as shown in Figure 4(h) (more details can be found in Movies S1 and S2). Finger-written patterns were directly mapped to control commands, including on/off switching, brightness adjustment, color temperature selection, and SOS signaling. The results validate real-time translation of user input into device action, proving the feasibility of our neuromorphic tactile system as wearable, lightweight, and low-power interfaces for interactive electronics. The same framework can be extended to extended reality/virtual reality (XR/VR) applications, where natural, screen-free interactions are highly desirable. By enabling haptic control without visual feedback or external sensors, our system provides an efficient neuromorphic tactile interface for human–machine communication.

4. Conclusion

Inspired by rodent vibrissae, we developed a wearable neuromorphic artificial fingernail that established a near-sensor computing framework for intelligent vibrotactile perception. Integrating piezoelectric sensing, spike encoding, and synaptic processing into a single platform enables hardware-level feature extraction and energy-efficient recognition with high accuracy. Compared with conventional digital circuits, our system reduces data complexity and energy consumption by several orders of magnitude. This efficiency arises from the event-driven spike representation and the nonlinear processing capability of the dual-gate synaptic transistor, which together enable in situ dynamic weighting and spatiotemporal integration.

Beyond vibrotactile sensing, our system supports diverse tactile tasks, including vibration, contour, and texture recognition, as well as tactile finger-writing-based interaction. This adaptability arises from modality-specific FA/SA encoding and asymmetric synaptic weighting, which reliably capture both dynamic and static features of tactile stimuli. Overall, this work establishes a biologically inspired, energy-efficient paradigm for wearable neuromorphic tactile perception, with promising applications in robotics, prosthetics, and human–machine interaction.

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Conflict of Interest

The authors declare that they have no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

[1] Mao Q, Liao Z J, Liu S Q, Yuan J F and Zhu R. 2025. An ultralight, tiny, flexible six-axis force/torque sensor enables dexterous fingertip manipulations. *Nat. Commun.* **16**, 5963.

[2] Ren H, Yang L, Chang H Y, Zhang T S, Li G, Yang X, Tang Y F, Shang W F and Shen Y J. 2025. A robust and omnidirectional-sensitive electronic antenna for tactile-induced perception. *Nat. Commun.* **16**, 3135.

[3] Xu M Z, Zhang J Q, Dong C Q, Tang C Y, Hu F X, Malliaras G G and Occhipinti L G. 2025. Simultaneous isotropic omnidirectional hypersensitive strain sensing and deep learning-assisted direction recognition in a biomimetic stretchable device. *Adv. Mater.* **37**, 2420322.

[4] Yue O Y, Wang X C, Liu X H, Hou M D, Zheng M H, Wang Y Y and Cui B Q. 2021. Spider-web and ant-tentacle doubly bio-inspired multifunctional self-powered electronic skin with hierarchical nanostructure. *Adv. Sci.* **8**, 2004377.

[5] Zhang L *et al*. 2024. Skin-inspired, sensory robots for electronic implants. *Nat. Commun.* **15**, 4777.

[6] Collinson D W, Emnett H M, Ning J Q, Hartmann M J Z and Brinson L C. 2021. Tapered polymer whiskers to enable three-dimensional tactile feature extraction. *Soft Robot.* **8**, 44-58.

[7] Lottem E and Azouz R. 2009. Mechanisms of tactile information transmission through whisker vibrations. *J. Neurosci.* **29**, 11686-11697.

[8] Petschenig H, Bisio M, Maschietto M, Leparulo A, Legenstein R and Vassanelli S. 2022. Classification of whisker deflections from evoked responses in the somatosensory barrel cortex with spiking neural networks. *Front. Neurosci.* **16**, 838054.

[9] Genc B, Ma L and Erzurumlu R S. 2001. Whisker-related neural patterns develop normally despite severe whisker defects in *Msx2* knockout mice. *Dev. Brain Res.* **132**, 107-111.

[10] Ego-Stengel V, Abbasi A, Larroche M, Lassagne H, Boubenec Y and Shulz D E. 2019. Mechanical coupling through the skin affects whisker movements and tactile information encoding. *J. Neurophysiol.* **122**, 1606-1622.

[11] Sehara K, Bahr V, Mitchinson B, Pearson M J, Larkum M E and Sachdev R N S. 2019. Fast, flexible closed-loop feedback: tracking movement in "real-millisecond-time". *eNeuro* **6**, 0147-19.

[12] Boubenec Y, Shulz D E and Debrégeas G. 2012. Whisker encoding of mechanical events during active tactile exploration. *Front. Behav. Neurosci.* **6**, 74.

[13] Wan C *et al*. 2018. An artificial sensory neuron with tactile perceptual learning. *Adv. Mater.* **30**, 1801291.

[14] Yang S M, Gao T, Wang J, Deng B, Lansdell B and Linares-Barranco B. 2021. Efficient spike-driven learning with dendritic event-based processing. *Front. Neurosci.* **15**, 601109.

[15] Ju F and Ling S F. 2014. Bioinspired active whisker sensor for robotic vibrissal tactile sensing. *Smart Mater. Struct.* **23**, 125003.

[16] Wegiriya H, Herzig N, Abad S A, Hadi Sadati S M and Nanayakkara T. 2020. A stiffness controllable multimodal whisker sensor follicle for texture comparison. *IEEE Sens. J.* **20**, 2320-2328.

Accepted Manuscript

[17] Liu Z *et al*. 2024. A three-dimensionally architected electronic skin mimicking human mechanosensation. *Science* **384**, 987-994.

[18] Wang W C *et al*. 2023. Neuromorphic sensorimotor loop embodied by monolithically integrated, low-voltage, soft e-skin. *Science* **380**, 735-742.

[19] Xu C H, Solomon S A and Gao W. 2023. Artificial intelligence-powered electronic skin. *Nat. Mach. Intell*. **5**, 1344-1355.

[20] Pammer L, O'Connor D H, Hires S A, Clack N G, Huber D, Myers E W and Svoboda K. 2013. The mechanical variables underlying object localization along the axis of the whisker. *J. Neurosci*. **33**, 6726-6741.

[21] Ralston B N, Flagg L Q, Faggin E and Birmingham J T. 2016. Incorporating spike-rate adaptation into a rate code in mathematical and biological neurons. *J. Neurophysiol*. **115**, 2501-2518.

[22] Shetty P, Shoykhet M and Simons D J. 2003. Whisker plucking alters responses of rat trigeminal ganglion neurons. *Somatosens. Mot. Res*. **20**, 233-238.

[23] Cui H Y *et al*. 2025. A bioinspired in-materia analog photoelectronic reservoir computing for human action processing. *Nat. Commun*. **16**, 2263.

[24] Zhang X M *et al*. 2020. An artificial spiking afferent nerve based on Mott memristors for neurorobotics. *Nat. Commun*. **11**, 51.

[25] Jiang C P, Liu J Q, Ni Y, Qu S D, Liu L, Li Y, Yang L and Xu W T. 2023. Mammalian-brain-inspired neuromorphic motion-cognition nerve achieves cross-modal perceptual enhancement. *Nat. Commun*. **14**, 1344.

[26] Zahid M, Gong J Y, Sadiq M I, Wang J W, Aslam F, Jin C X, Liu W R, Xu Y C, Yang J L and Sun J. 2025. Birdlike multisensory integrated oxide transistors for motion perception. *Nano Energy* **140**, 111045.

[27] Wang Y *et al*. 2021. MXene-ZnO memristor for multimodal in-sensor computing. *Adv. Funct. Mater*. **31**, 2100144.

[28] Chen S *et al*. 2024. Artificial organic afferent nerves enable closed-loop tactile feedback for intelligent robot. *Nat. Commun*. **15**, 7056.

[29] Luo Y Y *et al*. 2024. Adaptive tactile interaction transfer via digitally embroidered smart gloves. *Nat. Commun*. **15**, 868.

[30] Hyun W J, Secor E B, Kim C H, Hersam M C, Francis L F and Frisbie C D. 2017. Scalable, self-aligned printing of flexible graphene micro-supercapacitors. *Adv. Energy Mater*. **7**, 1700285.

[31] Liu Y H, Zhu L Q, Feng P, Shi Y and Wan Q. 2015. Freestanding artificial synapses based on laterally proton-coupled transistors on chitosan membranes. *Adv. Mater*. **27**, 5599-5604.

[32] Choi H W, Song K W, Kim S H, Nguyen K T, Eadi S B, Kwon H M and Lee H D. 2022. Zinc oxide and indium-gallium-zinc-oxide bi-layer synaptic device with highly linear long-term potentiation and depression characteristics. *Sci. Rep*. **12**, 1259.

[33] John R A, Ko J, Kulkarni M R, Tiwari N, Chien N A, Ing N G, Leong W L and Mathews N. 2017. Flexible ionic-electronic hybrid oxide synaptic TFTs with programmable dynamic plasticity for brain-inspired neuromorphic computing. *Small* **13**, 1701193.

[34] Long X Y, Deng B, Young C K, Liu G L, Zhong Z Q, Chen Q, Yang H, Lv S Q, Chen Z S and Zhang S J. 2022. Sharp tuning of head direction and angular head velocity cells in the somatosensory cortex. *Adv. Sci*. **9**, 2200020.

[35] Niu H S, Li H, Gao S, Li Y, Wei X, Chen Y K, Yue W J, Zhou W J and Shen G Z. 2022. Perception-to-cognition tactile sensing based on artificial-intelligence-motivated human full-skin bionic electronic skin. *Adv. Mater*. **34**, 2202622.

[36] Han J K, Tcho I W, Jeon S B, Yu J M, Kim W G and Choi Y K. 2022. Self-powered artificial mechanoreceptor based on triboelectrification for a neuromorphic tactile system. *Adv. Sci*. **9**, 2105076.

[37] Tashakori A *et al*. 2024. Capturing complex hand movements and object interactions using machine learning-powered stretchable smart textile gloves. *Nat. Mach. Intell*. **6**, 106-118.

[38] Ahmed A, Voelcker B and Peron S. 2024. Representational drift in barrel cortex is receptive field dependent. *Curr. Biol*. **34**, 5623-5634.e4.

[39] Diamond M E, Von Heimendahl M, Knutsen P M, Kleinfeld D and Ahissar E. 2008. 'Where' and 'what' in the whisker sensorimotor system. *Nat. Rev. Neurosci*. **9**, 601-612.

[40] Gerdjikov T V, Bergner C G, Stüttgen M C, Waiblinger C and Schwarz C. 2010. Discrimination of vibrotactile stimuli in the rat whisker system: behavior and neurometrics. *Neuron* **65**, 530-540.

Accepted Manuscript

[41] Haidarliu S, Nelinger G, Gantar L, Ahissar E and Saraf-Sinik I. 2024. Functional anatomy of mystacial active sensing in rats. *Anat. Rec.* **307**, 442-456.
[42] Wolfe J, Mende C and Brecht M. 2011. Social facial touch in rats. *Behav. Neurosci.* **125**, 900-910.
[43] Sun Z D, Zhu M L, Shan X C and Lee C. 2022. Augmented tactile-perception and haptic-feedback rings as human-machine interfaces aiming for immersive interactions. *Nat. Commun.* **13**, 5224.
[44] Yan Z G *et al*. 2021. Flexible high-resolution triboelectric sensor array based on patterned laser-induced graphene for self-powered real-time tactile sensing. *Adv. Funct. Mater.* **31**, 2100709.

Accepted Manuscript

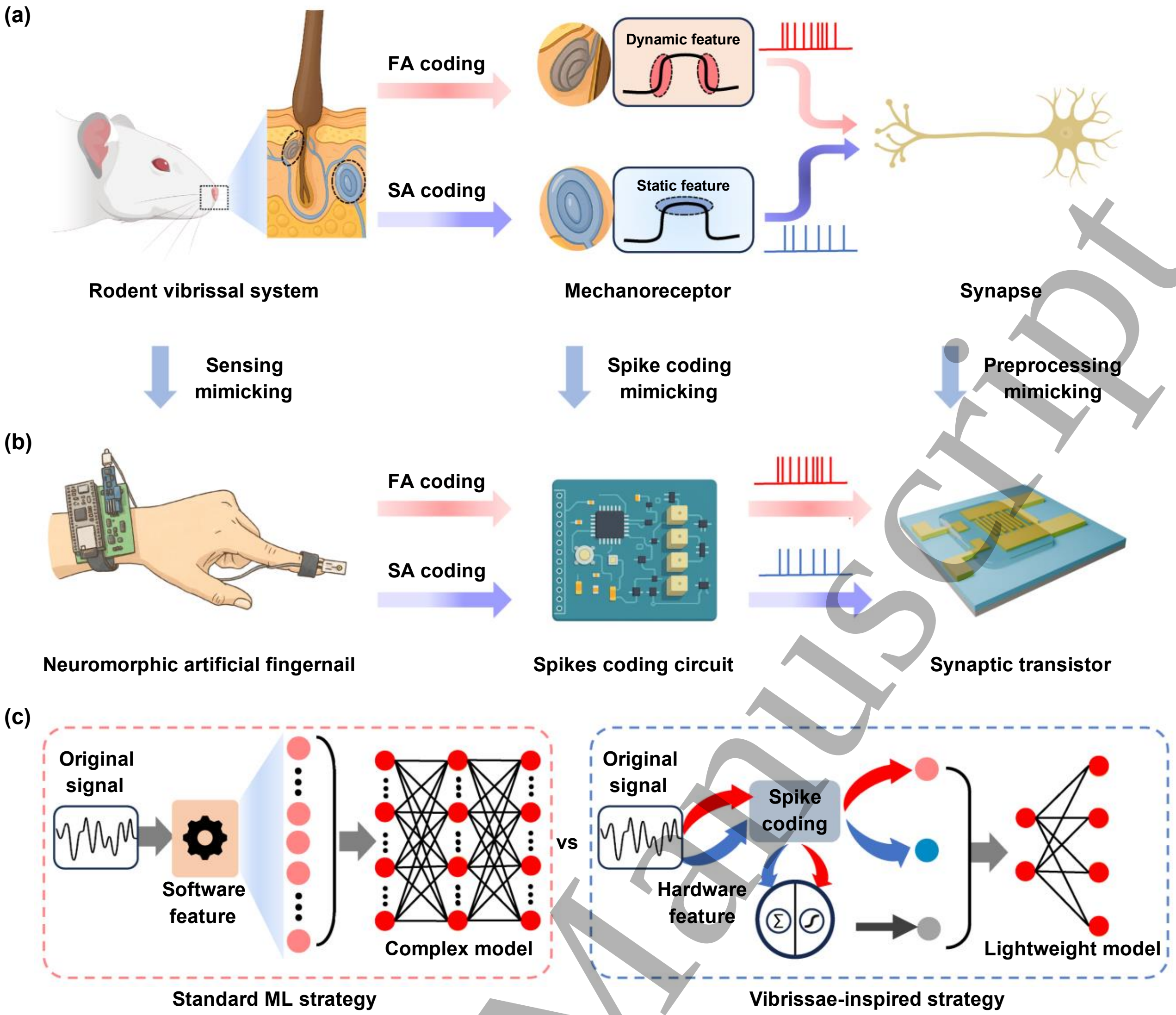

Figure 1. Bioinspired neuromorphic vibrotactile system. (a) Rodent vibrissal system where mechanoreceptors encode dynamic and static features that are integrated at the synapse. (b) Neuromorphic artificial fingernail-replicating sensing, spike coding, and synaptic preprocessing via a piezoelectric sensor, dual-channel encoder, and synaptic transistor. (c) Conceptual comparison between standard machine-learning pipelines requiring software feature extraction and complex models and the proposed vibrissae-inspired strategy that directly encodes hardware features for lightweight recognition.

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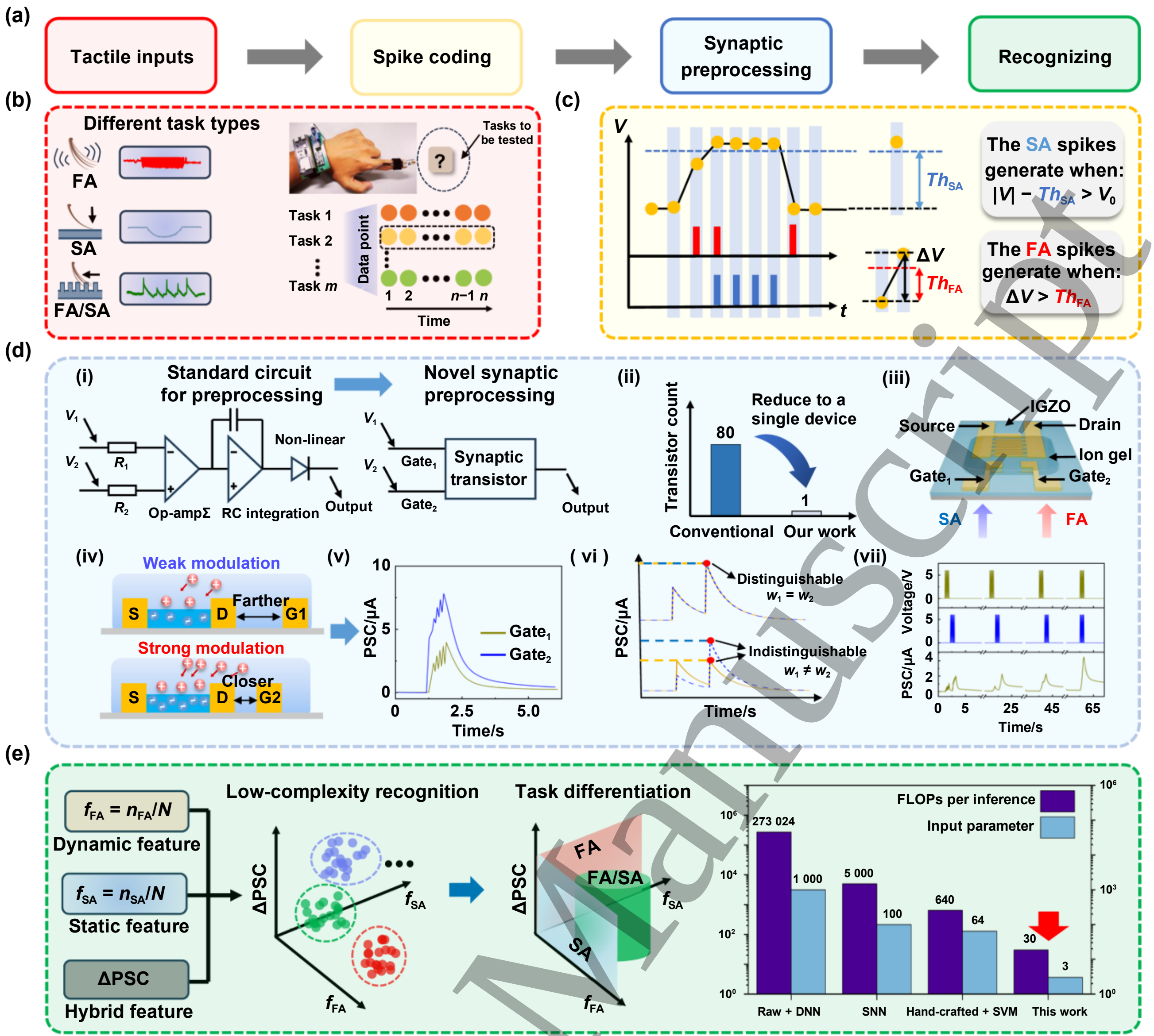

Figure 2. Neuromorphic tactile sensing and recognition framework. (a) Overall workflow of the neuromorphic tactile system. (b) Tactile inputs from vibration, pressure, and texture captured by a finger-mounted piezoelectric sensor. (c) FA/SA spike encoding scheme for converting continuous signals into sparse spike trains. (d) Synaptic preprocessing with a dual-gate transistor, showing reduced device count, asymmetric gate coupling, and spatiotemporal recognition capability. (e) Feature extraction and recognition using hardware-derived features, with reduced computational complexity compared with neural networks.

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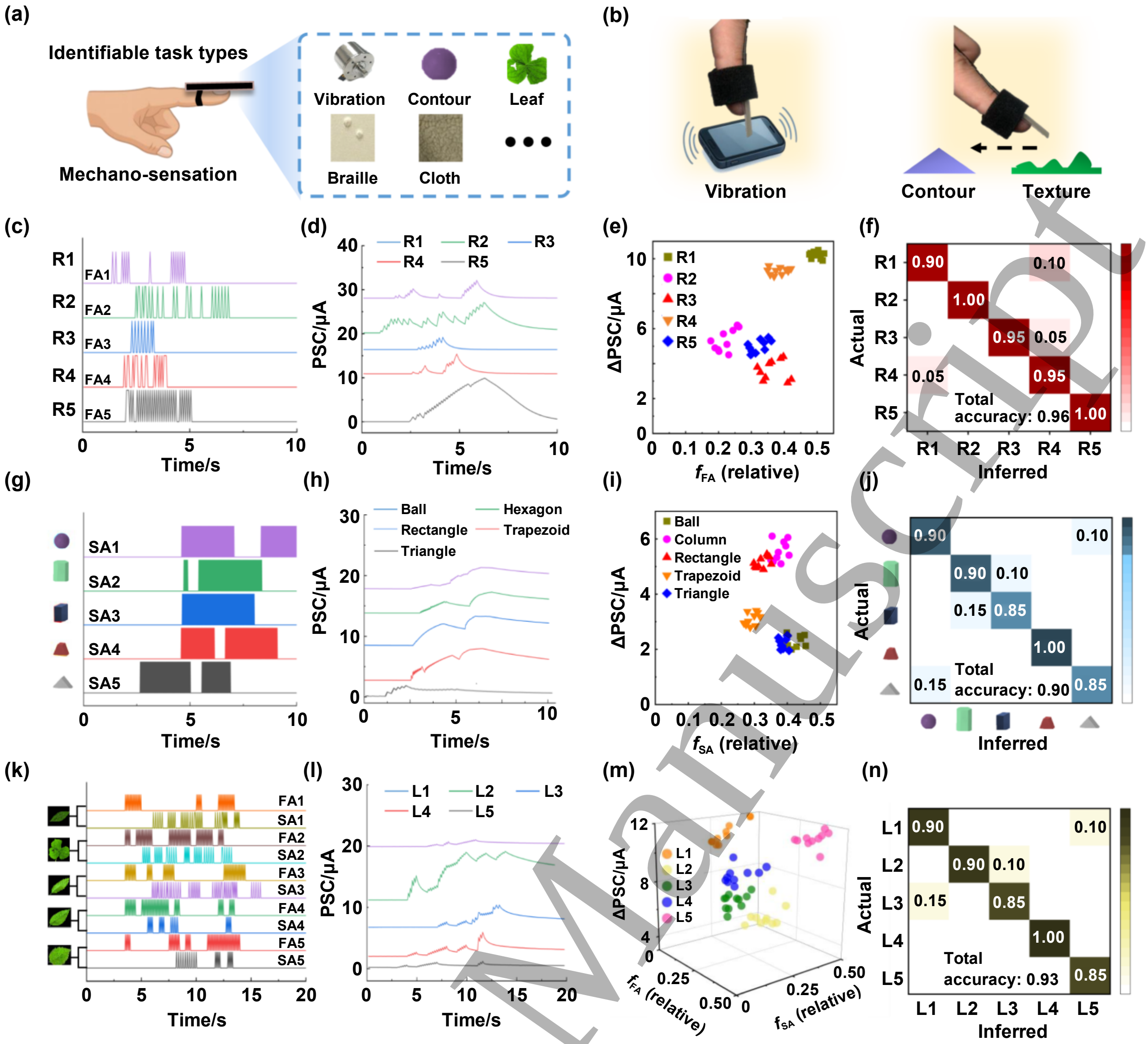

Figure 3. Dual-channel encoding enables the extraction and classification of diverse tactile features. (a) Task adaptability of the neuromorphic fingernail for vibration, contour, and texture perception. (b) Experimental setups for smartphone vibration sensing and contour/texture scanning. (c) FA-dominant spike trains generated by five vibration patterns. (d) Corresponding PSC responses under vibration inputs. (e) Feature distribution of vibration modes in the f_{FA}-ΔPSC plane. (f) Confusion matrix of vibration recognition. (g) SA-rich spike trains elicited when geometric contours are scanned. (h) PSC responses for different shapes. (i) Feature distribution of contour classes in the f_{SA}-ΔPSC plane. (j) Confusion matrix of contour recognition. (k) Spike trains generated during leaf-texture scanning. (l) PSC responses for different leaves. (m) Three-dimensional feature distribution (f_{FA}, f_{SA}, ΔPSC) for texture inputs. (n) Confusion matrix of texture classification.

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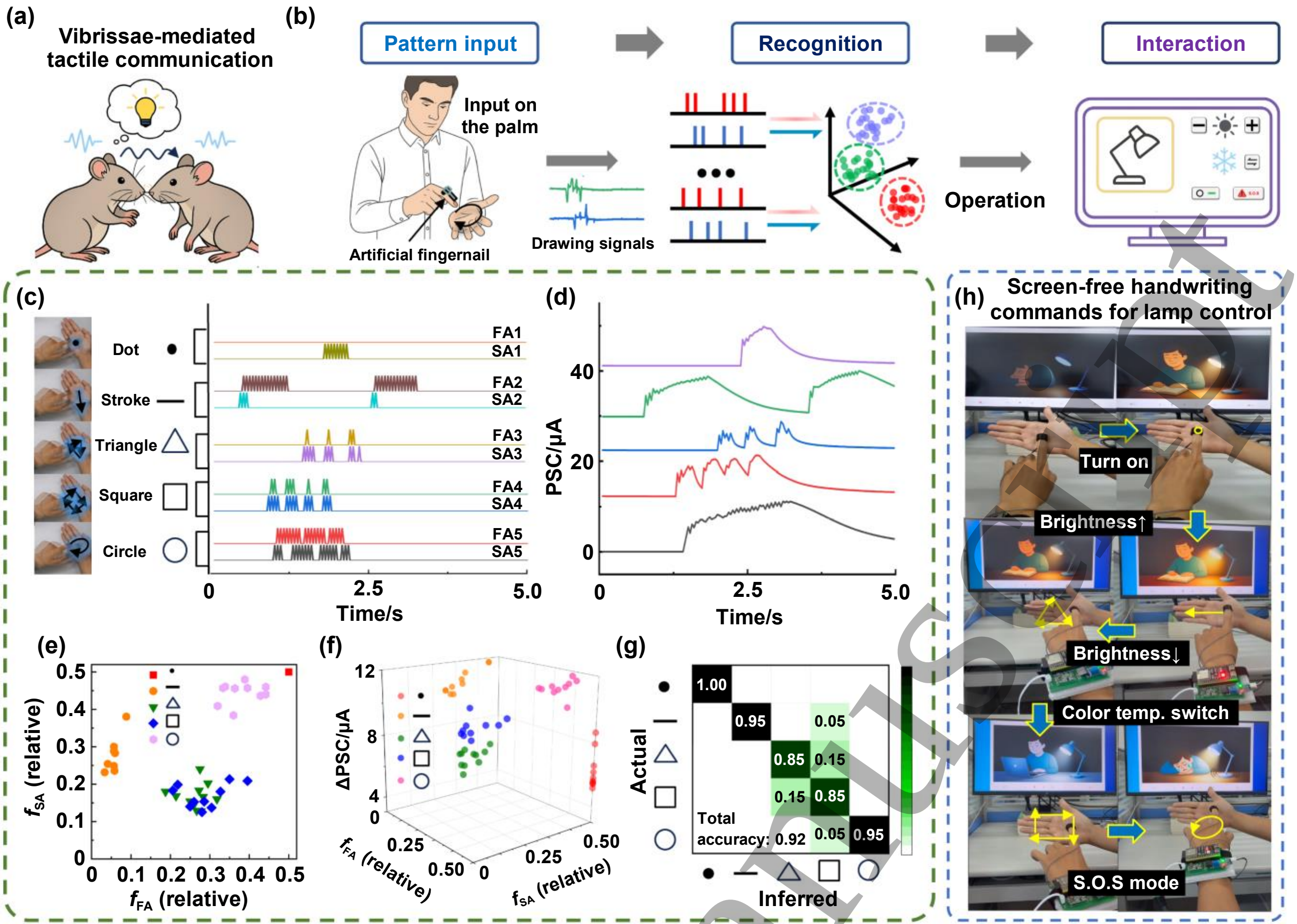

Figure 4. Neuromorphic tactile interface for real-time gesture recognition and control. (a) Bioinspired concept of vibrissal communication in rodents. (b) Workflow of the neuromorphic HCI framework for palm-written input and recognition. (c) FA/SA spike trains generated by five finger-written patterns. (d) Distinct PSC responses corresponding to different patterns. (e) Feature distributions in the f_{FA}-f_{SA} plane. (f) Enhanced separability in the 3D feature space (f_{FA}, f_{SA}, ΔPSC). (g) Confusion matrix showing classification accuracy above 90%. (h) Demonstration of palm-based finger-writing recognition for lamp control.

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